

COMPUTER HARDWARE

ZERO TO HERO

Basic Hardware • Components • Ports • Buttons • Troubleshooting

Entry-Level IT Support / Hardware Fundamentals

CORE COMPUTER COMPONENTS

1. Motherboard

Main circuit board connecting major computer components.

2. CPU

Processor that executes instructions and performs calculations.

3. RAM

Temporary working memory used by active programs and data.

4. Northbridge

Historical chipset component for high-speed communication with CPU, RAM, and graphics.

5. Southbridge

Historical chipset component for slower I/O and peripheral communication.

6. BIOS

Firmware that initializes hardware and starts the operating-system boot process.

7. UEFI

Modern firmware that initializes hardware and provides a modern boot environment.

8. CMOS Battery

Maintains firmware settings such as date, time, and configuration information.

STORAGE, GRAPHICS & CONNECTIONS

9. PSU

Power Supply Unit that converts AC power into regulated DC power.

10. HDD

Magnetic storage using spinning platters and mechanical read/write heads.

11. SSD

Flash-based storage with no mechanical moving parts.

12. GPU

Processes graphics and video workloads and sends output to a display.

13. CPU Fan / Heatsink

Removes processor heat and helps maintain safe operating temperatures.

14. PCIe Slot

Connects high-speed expansion devices such as graphics or network cards.

15. SATA Port

Connects compatible SATA storage devices to the motherboard.

16. M.2 Slot

Compact motherboard connector commonly used for SSDs and other supported devices.

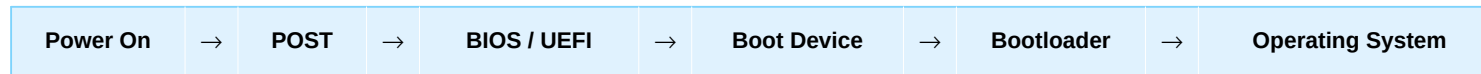
COMPUTER PORTS & BUTTONS

Port	Basic purpose
USB / USB-C	General peripheral and data connection
HDMI	Digital display connection
DisplayPort	Digital display connection
Ethernet / RJ-45	Wired network connection
3.5 mm Audio	Audio connection
VGA	Legacy analog display connection
PS/2	Legacy keyboard / mouse connection

Button	Basic purpose
Power Button	Turns the computer on or initiates the configured power action.
Reset Button	Restarts the system on hardware that provides a reset button.
Volume Buttons	Increase or decrease audio volume on supported devices.
Monitor Power	Turns the display on or off.

POST & BOOT PROCESS

POST (Power-On Self-Test) checks basic hardware before the operating system loads.



BASIC TROUBLESHOOTING

Problem	First checks
No Power	Check wall power/cable; PSU switch; motherboard & CPU power connectors; power-button connection.
No Display	Check monitor power/input; display cable; correct GPU/motherboard output; RAM; POST indicators.
Slow Computer	Check CPU/RAM/disk usage; free storage; startup/background applications; temperatures and cooling.
Overheating	Check fans; clean dust; check heatsink installation and airflow; check temperatures.
USB Not Working	Try another USB port; test the device on another computer; check cable; check drivers/device settings.

ENTRY-LEVEL HARDWARE INTERVIEW QUESTIONS

Q: What is a motherboard?

A: The main circuit board connecting major computer components.

Q: What is RAM?

A: Temporary working memory used by the computer for active programs and data.

Q: HDD vs SSD?

A: HDD uses magnetic mechanical storage; SSD uses flash memory.

Q: What is POST?

A: Power-On Self-Test; a startup check of basic hardware.

Q: What is a PSU?

A: Power Supply Unit; it provides regulated power to computer components.

Q: What is Northbridge?

A: A historical chipset component responsible for high-speed communication with CPU, RAM, and graphics.

Q: What is Southbridge?

A: A historical chipset component responsible for slower I/O and peripheral communication.

Q: First checks for no display?

A: Monitor power, input, cable, correct video output, RAM, and POST indicators.

QUICK REVISION SHEET

Motherboard	Main circuit board
CPU	Processes instructions
RAM	Temporary working memory
HDD	Magnetic storage
SSD	Flash storage
GPU	Graphics processing
PSU	Provides regulated power
BIOS / UEFI	Firmware and hardware initialization
CMOS Battery	Maintains firmware settings
Northbridge	Historical high-speed chipset function
Southbridge	Historical I/O chipset function
POST	Startup hardware test
USB	General peripheral/data connection
HDMI / DisplayPort	Digital display connections
RJ-45	Ethernet network connection
SATA	Storage interface
M.2	Compact motherboard connector

Thank You